

Project Proposal

● Graded

Group

Ammar Syed

Ali Hasan

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Title: Invisible Watermarking for Large Language Models via Logit Manipulation**Problem Statement**

As large language models (LLMs) become increasingly capable and widely deployed, the ability to reliably identify AI generated text is critical for mitigating misinformation, enforcing academic integrity, and maintaining trust in digital communications and media. Existing detection methods are focused on analyzing text after generation, such as perplexity based classifiers, quickly degrade under paraphrasing, minor edits, and unique or informal tones. A more robust approach would be to embed an imperceptible signal directly into the generation process, allowing attribution to survive the actions that are able to “trick” existing detection methods. The technical challenge in this project is striking a balance between competing goals. The signal needs to be strong enough to be able to be detected reliably but subtle enough to preserve text output quality. It should also be resilient to methods of post generation modifications such as substitutions, paraphrasing, and abbreviations. This challenge will be particularly difficult for shorter pieces of text, due to fewer tokens.

Related Work

Kirchenbauer et al. (2023) — *A Watermark for Large Language Models*. This work introduces the core logit-manipulation scheme we build upon. At each step of the decoding process, the vocabulary is split into green and red token lists using a pseudorandom hash of the previous token, and a bias δ is added to green-list logits before sampling. A detector then identifies watermarked text by checking whether green tokens appear at a statistically unusual rate, using a z-test. The main limitation is that the bias distorts the output distribution, which can degrade text quality at higher δ values, and detection becomes unreliable for very short passages. Our project builds directly on this scheme and extends its evaluation to short-text and paraphrasing settings.

Kuditipudi et al. (2023) — *Robust Distortion-free Watermarks for Language Models*. Rather than biasing the logit distribution, this paper maps a pseudorandom key sequence directly to token samples using inverse transform or exponential minimum sampling, leaving the output distribution completely unchanged. This makes it appealing from a quality standpoint, but the tradeoff is that detection is more involved. It requires aligning the observed text back to the original key sequence rather than simply counting green tokens. The authors demonstrate reliable detection from as few as 35 tokens even after 40–50% of tokens have been corrupted. We use this work as a conceptual and empirical counterpoint to our logit-bias approach when evaluating the tradeoff between output quality and detectability.

Kirchenbauer et al. (2024) — *On the Reliability of Watermarks for Large Language Models*. This follow-up paper tests how well watermarks actually hold up against human rewriting, LLM paraphrasing, and dilution into longer human-written documents. They find that even after paraphrasing, enough watermarked fragments survive, that detection remains viable given sufficient text. The limitation is that reliability drops considerably for short texts, and a motivated adversary using a spring paraphrase can still reduce detectability meaningfully. Our evaluation design draws directly from this work, replicating and extending several of its robustness tests.

Methodology**Dataset**

We will generate a text corpus by prompting target models with 400-600 or so prompts sampled from three public datasets: CNN/Daily Mail for summarization, WritingPrompts for open ended generation, and TriviaQA for question answering. This should give diversity across topics, style and tone, and output lengths. To evaluate quality, we'll measure perplexity via an independent reference model that is not used for generation. All the models and datasets are available on HuggingFace.

Architecture/Algorithms

We will implement a logit level watermarking scheme based on *Kirchenbauer et al. (2023)*. At each generation step, the prior token seeds a keyed hash function (SHA-256) that will partition the vocabulary into a green list with a green fraction and a red list. A fixed bias delta will be added to the

Ali Hasan (ah2434)

Ammar Syed (as4422)

green list logits before softmax and sampling. This should subtly cause the model to lean toward green list tokens without modifying the weights of the model. We can implement this as a HuggingFace LogitsProcessor and target two open source LLMS that have distinct architecture, Gemma3 4B and LLaMA 3.1 8B.

For detection, we rederive the green and red partition for each token using the same key and count how many fall into the green list. Text without a watermark should contain roughly 50% green list tokens by chance. Watermarked text will contain more, since the model was biased toward the green list tokens during generation. We will apply a one sided z test to determine whether the difference is statistically significant and report detection at a fixed false positive rate.

As a stretch goal, we will implement the semantic grouping variant from *Christ et al. (2024)*, which partitions tokens by meaning rather than randomly, and compare the tradeoffs between the two approaches.

Evaluation

In order to evaluate the model's performance, we will first look at detectability. We will measure the true positive rate at a fixed 1% false positive rate across page length, paragraph length, and sentence length passages and plot the z score distributions for watermarked versus unwatermarked text. To assess robustness, we'll test whether detection holds after adversarial modifications such as random word substitutions, paraphrasing, and token insertion/deletion. To measure whether the watermark degrades output quality and to what extent, we will compare the perplexity of the watermarked versus unwatermarked outputs using an independent reference model, measuring how quality degrades as we increase the bias delta. We might also look into reporting MAUVE scores to capture distributional similarity between watermarked and human text.

Deliverables and Expected Results

Deliverables include a custom implementation of a logit level watermarking injection and detection pipeline, an evaluation harness that generates watermarked and a control corpus and reports detection curves and statistics, and a final report analyzing tradeoffs and our findings.

Drawing on results from *Kirchenbauer et al*, we expected the watermark to be reliably detectable for passes of 200 or more tokens (TPR > 95% at 1% FPR). Performance should degrade for shorter texts. We expect moderate robustness to random word substitution, up to about 15% token replacement. We also expect reduced but still measurable detectability when LLMs paraphrase. Perplexity should increase relative to unwatermarked text should be fairly small for moderate bias values, showing that the watermark preserves output quality for the most part.

Feasibility

This project is computationally feasible on free or low cost resources on Google collab. We also have access to a gaming PC with an advanced NVIDIA RTX GPU. Generation and detection is inference only, model training and fine tuning is not required. Both target models (Gemma and LLaMA) should be able to be run on a T4 GPU via google colab and detection and analysis are CPU only tasks. All the datasets and models are publicly available on HuggingFace and readily accessible. The project should be able to be completed within the course timeframe by being broken down into modular pieces that can be worked on in a parallel manner.

Workload Distribution

Ali will work on the watermark injection implementation, including the LogitsProcessor and its integration with Gemma3 4B and LLaMA 3.1 8B, as well as the corpus generation pipeline. Ammar will build the detection module, including the z-test scorer and threshold calibration, and lead the robustness evaluation. This will cover word substitution, paraphrasing, and token insertion and deletion experiments. Quality evaluation (perplexity, MAUVE), the stretch goal, and the final report writing will be shared between both of us.

References

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- [3] Kirchenbauer, J., Geiping, J., Wen, Y., Shu, M., Saifullah, K., Kong, K., Fernando, K., Saha, A., Goldblum, M., & Goldstein, T. (2024). On the reliability of watermarks for large language models. *ICLR 2024*.
- [4] Christ, M., Gunn, S., & Zamir, O. (2024). Undetectable watermarks for language models. *COLT 2024*.